

Why is the boomerang coming back?

X18 (2018) — English translation

In this problem we will study the movement of a boomerang. Its exact movement is difficult to describe, but with certain assumptions we can understand why the boomerang returns to the point of throwing. Consider a symmetrical cross-shaped boomerang with a uniform mass distribution. The total mass of the boomerang is M , the length and width of the blades are R and $a \ll R$, respectively. The thickness of the boomerang can be neglected in comparison with these values. The boomerang is thrown so that its plane is vertical. As a result of the throw, the boomerang begins to rotate with an angular velocity of ω relative to the axis of symmetry perpendicular to the boomerang plane. The center of mass of the boomerang begins to move with a horizontal speed v , lying in the plane of the boomerang. During movement, a hydrodynamic force acts on the boomerang blades. This force is perpendicular to the plane of the blades. If the boomerang rotates as shown in the figure, then this force is directed away from us. The hydrodynamic force acting on a small element of the blade area ΔA is equal to

$$F = \gamma v_{\perp}^2 \Delta A.$$

где $v_{\perp}$ — компонента скорости воздуха относительно этого элемента площади, направленная перпендикулярно краю лопасти. Постоянная γ is proportional to the air density (which is considered constant) and depends on the shape of the boomerang. The occurrence of this hydrodynamic force is due to a slight bending of the surface of the boomerang, as a result of which the air incident on its surface acts on the surface in a direction perpendicular to it. The bending of the boomerang surface can be neglected within the framework of this problem. You can also neglect gravity and air resistance.

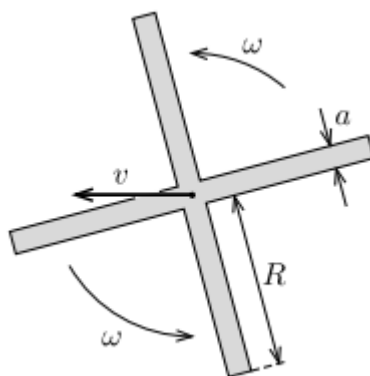


Fig. 1.

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| A1 | Find the total hydrodynamic force $\langle F \rangle$ acting on the boomerang, time-averaged per revolution. Express your answer in terms of v , ω , R , a and γ . | 2.50 |
| A2 | Find the total moment of hydrodynamic forces $\langle \vec{\tau} \rangle$ acting on the boomerang relative to its center. The torque must be averaged over time per revolution. Express your answer in terms of v , ω , R , a and γ . Indicate the direction of the average moment relative to the direction of the velocity vector of its center. | 2.50 |

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| A3 | What should the ratio $v/(\omega R)$ be equal to at the moment of throwing the boomerang, so that during the subsequent movement the center of the boomerang moves along a horizontal circular path? Consider that the period of rotation of the boomerang $2\pi/\omega$ is much less than the period of circular motion of the center of mass. This allows us to use the values $\langle F \rangle$ and $\langle \vec{\tau} \rangle$ averaged over the period found earlier. | 3.00 |
| A4 | Express the radius r of the trajectory of the boomerang's center of mass in terms of γ and the parameters of the boomerang. | 1.50 |
| A5 | Suppose we made two similar boomerangs from the same material, such that each size of one of them is half the corresponding size of the second. What is the ratio of the radii r_2/r_1 of their circular trajectories? | 0.50 |

Source: <https://pho.rs/p/316>